# SubCellSpace: A Unified Platform for Subcellular Spatial Transcriptomics Analysis

F. Li · School of Life Sciences, Tsinghua University · 2026

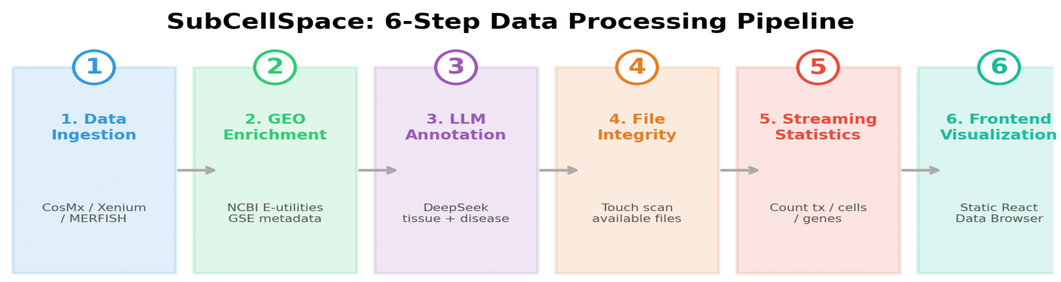
## 1. Project Overview

SubCellSpace is a comprehensive subcellular spatial transcriptomics platform integrating a 6-step scverse-native pipeline, a curated GEO database (2,131 entries), and a static web browser.

Supported Platforms:
- NanoString CosMx — High-plex in situ RNA detection
- 10x Genomics Xenium — Subcellular + multimodal imaging
- Vizgen MERFISH — Multiplexed FISH, ultra-high plexity

All processing unified under a single YAML configuration, with plugin-style backends for each pipeline step.

## 2. Analysis Pipeline

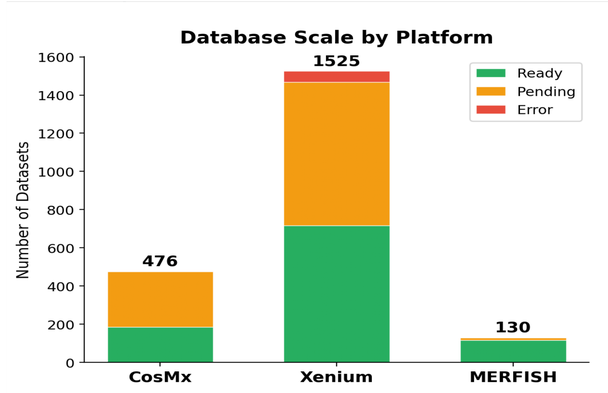


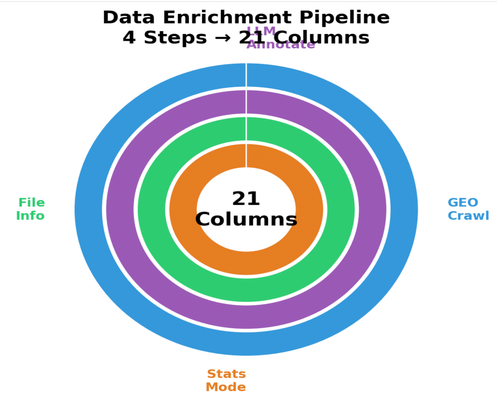

## 3. Pipeline Steps & Backends

Step 1 — Denoise: basic, sparc
Step 2 — Patchify: grid, hex
Step 3 — Segmentation: fov_cell_id, cellpose, baysor, proseg
Step 4 — Spatial Domain: spatial_leiden, spatial_kmeans
Step 5 — Clustering: leiden, louvain
Step 6 — Annotation: celltypist, scrin

Spatial Analysis — moran, geary (squidpy)
Subcellular Analysis — scrin (RNA localization)
Subcellular Domain — hdbscan, dbscan, phenograph

All backends registered via @register_backend decorator, selected through config/pipeline.yaml, fully data-driven.



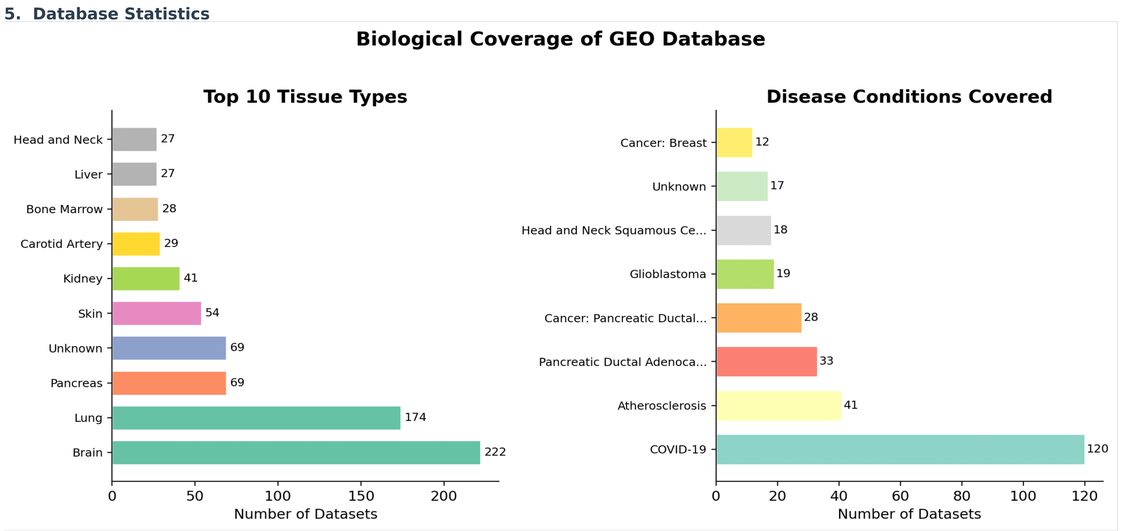



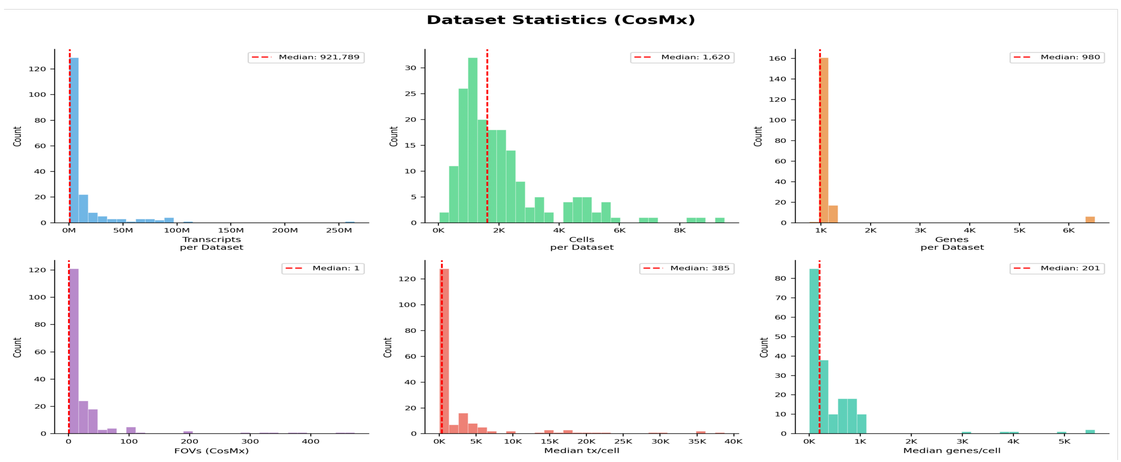

Dual database architecture:
- datasets.csv — Legacy non-GEO (998 rows)
- *_geo.csv — Per-platform GEO (2,131 rows)

4-step enrichment pipeline:
1. GEO Crawl — NCBI E-utilities → GSE metadata
2. LLM Annotation — DeepSeek → tissue, disease
3. File Integrity — Touch module → available files
4. Streaming Stats — tx / cells / genes counts

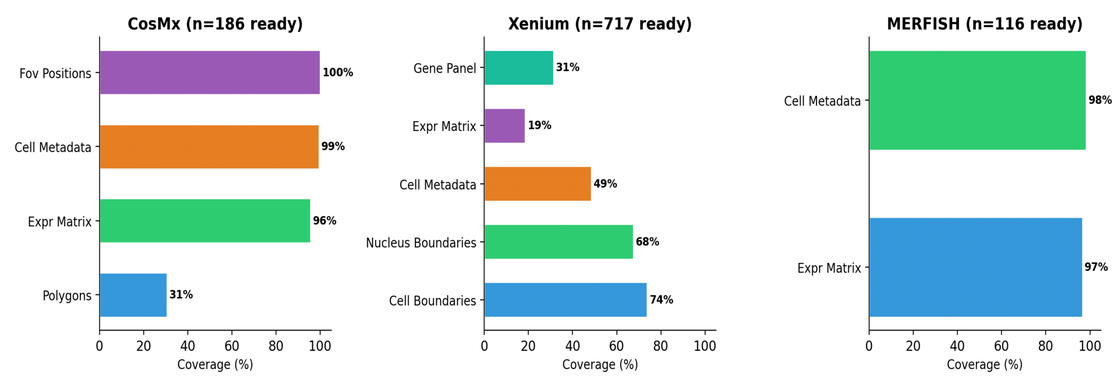
21 standardized columns per entry

## 5. Database Statistics




## 6. File Coverage & Frontend




Companion file types per platform:
CosMx: polygons, expr_matrix, cell_metadata, fov_positions
Xenium: cell_boundaries, nucleus_boundaries, expr_matrix (h5/mtx), cell_metadata, gene_panel
MERFISH: expr_matrix, cell_metadata

Frontend: Pure static React browser
- Platform tabs (CosMx / Xenium / MERFISH)
- 21-column sortable table with category headers
- Full-text search, status filtering
- GEO hyperlinks, row detail modals
- No backend required — static JSON data source

## 7. System Architecture & Tools

Core Stack:
Python 3.12 + scverse (scanpy, squidpy, spatialdata)
FastAPI server · React 18 + TypeScript + Vite
Slurm HPC job arrays for batch processing

Output Formats:
cells.parquet + metadata.json (container export)
subcellspace_report.html (interactive HTML report)
subcellspace_results.h5ad (full AnnData)
subcellspace_touch_report.json (file integrity)

External Tools:
SCRIN — Subcellular RNA localization & interactivity
SOPA — Segmentation pipeline (Cellpose, Baysor, etc.)

Infrastructure:
a-cluster (377GB RAM, 80 CPU, 218TB storage)
gpu01 — NVIDIA GPU node for segmentation
bio-download jumpbox for GEO data transfer

Data Processing:
3M+ transcripts streamed per dataset
Memory-safe streaming with 50k cell threshold
Incremental checkpoint saves for batch runs